MIL-F-51286B(EA)
8 August 1983
SUPERSEDING
MIL-F-51286A(EA)
9 February 1977

MILITARY SPECIFICATION

FACEBLANKS, CHEMICAL-BIOLOGICAL MASK, C17R1

This specification is approved for use by US Army Armament, Munitions and Chemical Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a chemical-biological mask faceblank molded from a 65-percent natural rubber compound.

1.2 Classification. The faceblank shall be of the following sizes, as specified (see 6.2):

Small (Drawing E5-2-1031)
Medium (Drawing E5-2-1032)
Extra Small (Drawing E5-2-1500)

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed: to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-TSC-S, Aberdeen Proving Ground, MD 21010 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4240

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STANDARDS

FEDERAL

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS

US ARMY ARMAMENT, MUNITIONS AND CHEMICAL COMMAND

CHEMICAL RESEARCH AND DEVELOPMENT CENTER

A5-2-794	-	Medallion.
D5-2-1027	-	Form, Faceblank, Small, Sections.
D5-2-1028	-	Form, Faceblank, Medium, Sections.
E5-2-1029	-	Form, Faceblank, Small.
E5-2-1030	-	Form, Faceblank, Medium.
E5-2-1031	-	Faceblank, Small.
E5-2-1032	-	Faceblank, Medium.
E5-2-1500	-	Faceblank, Extra Small, C17R1.
E5-2-1501	-	Form, Faceblank, Extra Small.
D5-2-1502	-	Form, Faceblank, Extra Small, Sections.
D136-19-14	-	Puncture Tester, Natural Rubber, Q38, Assembly.
E136-19-22	-	Flexibility Tester, Natural Rubber, Q37, Assembly.

PUBLICATION

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

CHEMICAL SYSTEMS LABORATORY DIRECTIVE

No. 618 - Ozone Resistance Test of Natural Rubber Material.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

ASTM STANDARDS

D412 - Tension Testing of Vulcanized Rubber.
D572 - Accelerated Aging of Vulcanized Rubber by the Oxygen-Pressure Method.
D573 - Accelerated Aging of Vulcanized Rubber by the Oven Method.
D624 - Tear Resistance of Vulcanized Rubber.
D2240 - Indentation Hardness of Rubber and Plastics by Means of a Durometer.

(Application for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103.)

SCHWARTZ, TULIPAN, AND BIRMINGHAM

Occupational Diseases of the Skin

(This publication may be obtained from the publishers: Lea and Febiger, Philadelphia, PA.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Materials.

3.1.1 Rubber compound. The rubber compound shall be nonblooming, except for a slight wax-type bloom, and shall contain no less than 65 percent, by volume, of new natural rubber. The compound shall be free from lead or mineral sulfates.

3.1.2 Antioxidants and accelerators. The rubber compound shall contain only approved antioxidants and accelerators (see 3.1.6).

3.1.3 Antiozonide. The rubber compound shall contain an antiozonide.

3.1.4 Copper and manganese inhibitor. The rubber compound shall contain a copper and manganese inhibitor.

3.1.5 Coloring materials. The rubber compound shall be colored by carbon black as required.

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3.1.6 Toxicity. When the faceblank is worn as a component of a chemical-biological mask, it shall not cause undue sensitization or primary irritation of the wearer's skin. Objective evidence shall be submitted in accordance with 4.1.2.2. Once the rubber formulation has been approved by the Government, the contractor shall not make a major change (6.4) in the formulation or in the curing process.

3.2 Manufacture. The faceblank shall be manufactured in one piece by the conventional process of molding under heat and pressure. Each size specified in the contract or order shall be manufactured in accordance with the applicable drawings listed in 2.1 in molds approved by the Government. There shall be no imperfections or damage in critical areas (see 3.5).

3.2.1 Repair of flaws. Faceblanks with small flaws (major or minor defects) may be repaired and accepted provided the repair is vulcanized with material of the same approved formulation as the faceblank and the repaired area meets the requirements of physical properties of 3.3. The appearance of the repair shall be such that the color and shape blend into the surrounding area with no break in surface continuity and no feather edges.

3.2.2 Test slabs. Test slabs 6 by 6 inches, by 0.075 $\pm$ 0.01 inch shall be made from stock representative of the formulation used for the preproduction sample and from each sublot during regular production of faceblanks. The slabs shall have a cure equivalent to that of the regular production faceblank (see 6.3).

3.3 Physical properties. Physical properties shall conform to the requirements shown in table I when tested as specified therein.

3.4 Preproduction. Prior to the start of regular production, a preproduction lot of faceblanks shall be produced for examination and testing in accordance with 4.3. Any change in formulation, manufacturing methods, or molds shall require a new preproduction lot.

3.5 Critical areas. The peripheral sealing area contacting the wearer's skin, the inner and outer surfaces of the filter element nozzles, the spectacle insert block, the sealing surface for the voicemitter-outlet valve assembly, and the lens sealing surface shall be considered as critical areas of the faceblank.

3.6 Workmanship. The faceblank inner and outer surfaces shall be smooth and free from imperfections such as pits, bumps, porous areas, cracks, flash, abrasions, checks, blisters, or foreign matter. The rubber shall be compact and free from holes and bubbles. Ridges of mold-parting lines on outer surfaces shall not be considered as flash when closely trimmed. Rips, tears, or punctures shall be considered as damage. Workmanship defects of a magnitude greater than the referee standards established by the Government for major and minor defects (see 6.7) shall be categorized in the appropriate defect class.

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3.7 Resistance to oxygen-pressure aging. The faceblank shall show no cracking or increased hardness or tackiness after 72 hours of accelerated aging in oxygen, at a pressure of 2.10 ± 0.10 megapascals (MPa) and a temperature of 70° ± 1°C, as specified in 4.4.4.2.5.

TABLE I. Physical properties

Property	Unaged	After heat aging*	Test paragraph	ASTM Test Method
Average tensile strength, MPa, min	20.7	17.6	4.4.4.2.1	D412
Average ultimate elongation, %, min	600		4.4.4.2.1	D412
Percent increase, max		20		
Percent decrease, max		25		
Average tensile stress at 200% elongation, MPa	1.72 to 3.10	1.72 to 3.10	4.4.4.2.1	D412
Individuals, MPa, min	1.65	1.65		
Average tear resistance, kN/m, min			4.4.4.2.1	D624
Die A or B	52.5	36.8		
Die C	39.4	28.9		
Average tension set at 75% ultimate elongation, %, max	20		4.4.4.2.1	D412
Average hardness, Shore "A" or "A2"	47 ± 3		4.4.4.2.1	D2240
Ozone resistance, min	80		4.4.4.2.2	
Flexibility			4.4.4.2.3	
Units at 24 ± 3°C, min	7.5			
Units retained at -32° ± 3°C min	4.8			
Puncture resistance, psi, min	9.5		4.4.4.2.4	

*Test specimens selected for heat aging shall be aged in an air oven at a temperature of 70° ± 1°C for a continuous period of 168 ± 1 hours or at a temperature of 100° ± 1°C for a continuous period of 24 ± 1 hours as specified in ASTM D573 (see 6.6).

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3.8 Lot marking. Lot number as specified by the procuring activity s embossed on the face blank as required by Drawing E5-1-553.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the tract or purchase order, the contractor is responsible for the performan all inspection requirements as specified herein. Except as otherwise sp in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements s fied herein, unless disapproved by the Government. The Government reser right to perform any of the inspections set forth in the specification w such inspections are deemed necessary to assure supplies and services co to prescribed requirements.

4.1.2 Objective evidence.

4.1.2.1 Materials. The contractor shall provide objective evidence a able to the contracting officer that the requirements of 3.1 and section which specific inspection has not been provided in this specification ha satisfied.

4.1.2.2 Toxicity. After date of award of contract, and within the ti specified in the contract, and without additional cost to the Government contractor shall submit the information listed below which will constitu evidence for establishing the nontoxicity of the formulation. The US Ar Surgeon General will act as advisor to the contracting officer in evalua this data for determining if the formulation and parts are acceptable. formation relating to the description of the candidate formulations may classified as "company confidential," in which case it will be so treate the Government.

a. Information relating to candidate formulations:

(1) The name of the contractor and the producing subcon tor, if any.

(2) Description of the item and end item and copies of applicable specifications shall be included.

(3) The number of items under procurement for which the posed formulation shall be used.

(4) A description of the candidate formulation shall in

(a) The chemical and trade names of the ingredient

(b) The quantity of each ingredient, including the of variation that can be tolerated without loss of quality in the finish item. Quantities shall be expressed in percent by weight and parts by w

(c) The proposed mold lubricants, dusting powders, spray or other materials used in the manufacturing process.

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(d) Proposed cure times and temperatures.

(e) Results of previous toxicity evaluation demonstrating acceptability or evidence of prior Government approval of the proposed formulation.

(f) Information and data to substantiate the non-toxicity of the product. This information will include a complete description of any programs the contractor has carried out to prove nontoxicity, such as prior patch testing with unlined items the use of which involves direct contact with the skin; evidence that the proposed formulation is the same as that used in his commercial products; and information on the nature, extent, and length of time of his commercial experience with the formulation. Other evidence may include references to technical or other scientific publications, commercial laboratory reports, opinions or statements of recognized authorities and agencies in the field of dermatology, and any other pertinent data.

(b) Patch testing. If the above information is not sufficient to establish nontoxicity of the rubber formulation, the contractor shall perform 200 patch tests under the supervision of competent medical authorities in the field of dermatology. These patch tests shall be at no additional cost to the Government and shall be conducted in accordance with the prophetic patch test method described in "Occupational Diseases of the Skin" by Schwartz, Tulipan, and Birmingham. A control rubber for use in the patch tests will be furnished by the Government upon written request of the contractor. The patches composed of the candidate rubber formulation shall not produce positive reactions in excess of that of the control rubber patches. A copy of the test results and the evaluation thereof by the medical authorities described above shall be furnished to the contracting officer.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 Preproduction inspection.

4.3.1 Sample. A preproduction sample of 30 (20 trimmed and 10 untrimmed) faceblanks from each mold cavity and 16 test slabs shall be manufactured using the same methods, materials, equipment, and rubber formulations as will be used during regular production.

4.3.2 Inspection procedure.

4.3.2.1 For examination. Fifteen of the sample faceblanks, from each mold cavity, shall be inspected for all requirements of the drawings and this specification.

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4.3.2.2 For tests. Fifteen faceblanks, selected randomly from the rem sample, and 16 test slabs, produced in accordance with 4.3.1, shall be te in accordance with 4.4.4.

4.3.3 Acceptance/rejection criteria. The preproduction sample shall m the examination and tests specified by 4.3.2.1 and 4.3.2.2 to be acceptal The contractor shall obtain written approval from the contracting officer before proceeding with regular production.

4.4 Quality conformance inspection.

4.4.1 Lotting. A lot shall consist of the faceblanks of one size proc in sequence during one month as indicated by the dots in the medallion or faceblank. A sublot shall consist of the faceblanks produced from one ba material (see 6.5). A new lot shall be established following the occurer any event affecting the homogeneity of the lot or use of an alternative approved formulation.

4.4.2 Sampling.

4.4.2.1 For examination and nondestructive tests. Sampling shall be conducted in accordance with MIL-STD-105.

4.4.2.2 For destructive tests. Fifteen faceblanks, selected randomly the faceblank sublot, and 16 tests slabs shall be submitted with each sub for testing.

4.4.3 Inspection procedure.

4.4.3.1 For examination and nondestructive tests. The sample faceblan from the regular production lot shall be examined in accordance with the classification of defects and with MIL-STD-105.

4.4.3.2 For destructive tests. Sample faceblanks and test slabs shall tested in accordance with 4.4.4. The acceptance number is zero.

4.4.3.3 Classification of defects.

(a) Faceblank. See applicable drawing for size.

Categories		Defects
Critical:		None defined
Major:	AQL	1.0 percent defective
101		Rubber thickness incorrect
102		Porosity or nonfills
103		Deformation or permanent set

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104	Flow cracks which enlarge on probing
105	Marking incorrect
106	Workmanship (other than Major 110, see 3.6)
107	Mismatched bag mandrels
108	Split at register
109	Spectacle insert incorrect

AQL 0.065 percent defective

110	Imperfections or damage in critical areas (see 3.5)
111	Multiple marking on single tab
112	Tackiness of rubber in lot marking area

Minor: AQL 2.5 percent defective

201	Excess flash or improper trim
202	Marking missing or illegible
203	Access flap holes improperly located
204	Lot marking in wrong location on forehead tabs

4.4.4 Tests.

4.4.4.1 Test specimens. Two specimens, one from each cheek, shall be cut from the flat cheek areas of each of 15 sample faceblanks. The specimens shall not be buffed or stretched before testing. Ten specimens, five from each cheek, shall be tested for tensile strength, ultimate elongation, and tensile stress before aging and 10 specimens tested after aging. Six specimens, three from each cheek, shall be tested for tension set before aging. Two specimens for tear resistance shall be cut from the forehead section of each of 10 of the above 15 sample faceblanks. Ten specimens shall be tested before aging and 10 tested after aging. Enough 1- by 1-inch squares shall be cut from cheek and forehead sections to build up five specimens for the hardness test, each specimen containing enough layers to be no less than 3/16 inch thick; these squares shall also be used (after the hardness test) in the test for resistance to oxygen-pressure aging. The remaining faceblanks shall be used for retest. Twelve specimens, two from each of six test slabs, shall be taken for ozone resistance tests. Eight specimens, two from each of four test slabs, shall be taken for flexibility testing. Six specimens, one from each of six test slabs, shall be taken for puncture resistance tests.

4.4.4.2 Physical properties tests.

4.4.4.2.1 Tensile strength, ultimate elongation, tensile stress, tear resistance, tension set, and hardness. Tensile strength, ultimate elongation, tensile stress, tear resistance, tension set, and hardness shall be determined in accordance with the ASTM standards, specified in table I. The angle-type or crescent-type die shall be used for tear resistance testing and specimens used for the hardness test shall be set aside for use in the test for resistance to oxygen-pressure aging.

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4.4.4.2.2 Ozone resistance. Ozone resistance shall be determined in accordance with Directive No. 618.

4.4.4.2.3 Flexibility. Flexibility shall be determined with the Q37 Natural Rubber Flexibility Tester Assembly (E136-19-22). Flexibility shall be read on the grid of the apparatus. Four specimens shall be tested at 24° ± 3°C and four specimens shall be tested at -32° ± 3°C after being conditioned for 72 hours at that temperature.

4.4.4.2.4 Puncture resistance. Resistance to hydrostatic pressure shall be determined with the Q38 Natural Rubber Puncture Tester Assembly (D136-19-14).

4.4.4.2.5 Resistance to oxygen-pressure aging. Rubber specimens from the faceblank shall be tested in accordance with ASTM D572.

5. PACKAGING

5.1 Unit packing, packing, and marking, interplant shipment (see 6.8). The faceblanks shall be unit packed and packed to provide adequate protection from physical damage from the supply source to the first receiving activity for immediate use or further processing. Shipping containers shall be in compliance with the rules and regulations applicable to the mode of transportation. Marking shall be in conformance with Fed. Std. No. 123. The following additional marking shall appear on all unit packs and shipping containers.

(1) Faceblank size
(2) Composition of faceblank and formulation code
(3) Nomenclature of mask to which faceblank pertains
(4) Date of cure.

6. NOTES

6.1 Intended use. The faceblanks covered by this specification are intended for use in the manufacture of M17A1 and M17A2 Field Chemical-Biological Masks.

6.2 Ordering data. Acquisition documents should specify:

(a) Title, number, and date of this specification.
(b) Size of faceblank.
(c) Applicable drawings.
(d) Preproduction:

(1) Time allowed for contractor submission of samples for Government test and evaluation after award of contract.
(2) Name and address of test facility and shipping instructions when testing is performed by the Government.
(3) Time required for the Government to notify the contractor whether or not to proceed with production.

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6.3 Equivalent cure. Slabs and faceblanks of the same stock and gage are considered to have equivalent cures or the same state of vulcanization when:

(a) Their free-sulfur contents do not vary more than 10 percent.
(b) Their average tensile stress values at any given elongation (other than ultimate elongation) do not vary more than ± 2.5 percent.
(c) Their curing times can be equated in the following equation:

$$t_2 = t_1 \times 1.6^{(T_1 - T_2)/10}$$

where: t_1 = minutes of cure at temperature T_1,
t_2 = minutes of cure at temperature T_2,
T_1 = temperature (°F) of cure for time t_1, and
T_2 = temperature (°F) of cure for time t_2.

For convenience, the nomograph in figure 1 may be used instead of the mathematical computation time-temperature relationship in determining equivalent cure.

6.4 Major change. The term applies to any change in formulae, processing, or curing of currently approved rubber stocks which induce toxic characteristics and cause casualties. The major change should be considered effected when one or more of the followng conditions occur:

(a) Components of formulae are varied in excess of the limitations stated in the manufacturer's tolerance (see 4.1.2.2).
(b) The addition or deletion of one or more components of formulae.
(c) Substitutes for one or more components of formulae without advance approval of the Government.
(d) Deviation from the submitted approved factory mold cure temperature exceeding +5°F or -10°F, even though equivalent cure is attained.
(e) Use of mold lubricants, dusting powders, or wet sprays not submitted with original information for toxicity approval.

6.5 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process or subjected to some physical mixing operation intended to make the final product substantially uniform.

6.6 Aging. Aging at 100° ± 1°C for a continuous period of 24 hours is slightly more severe on rubber than aging at 70° ± 1°C for 168 consecutive hours; therefore, specimens that fail to meet requirements when aged at the higher temperature may be discarded and replaced by new specimens aged at 70 ± 1°C for 168 consecutive hours.

6.7 Referee standards. During preproduction and subject to later revision during production, the Government will establish referee standards to define the type and category (major and minor) of each defect. This classification will be based upon the defect magnitude and potential deleterious effect on intendd use.

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EQUIVALENT CURE TIME FOR NATURAL RUBBER

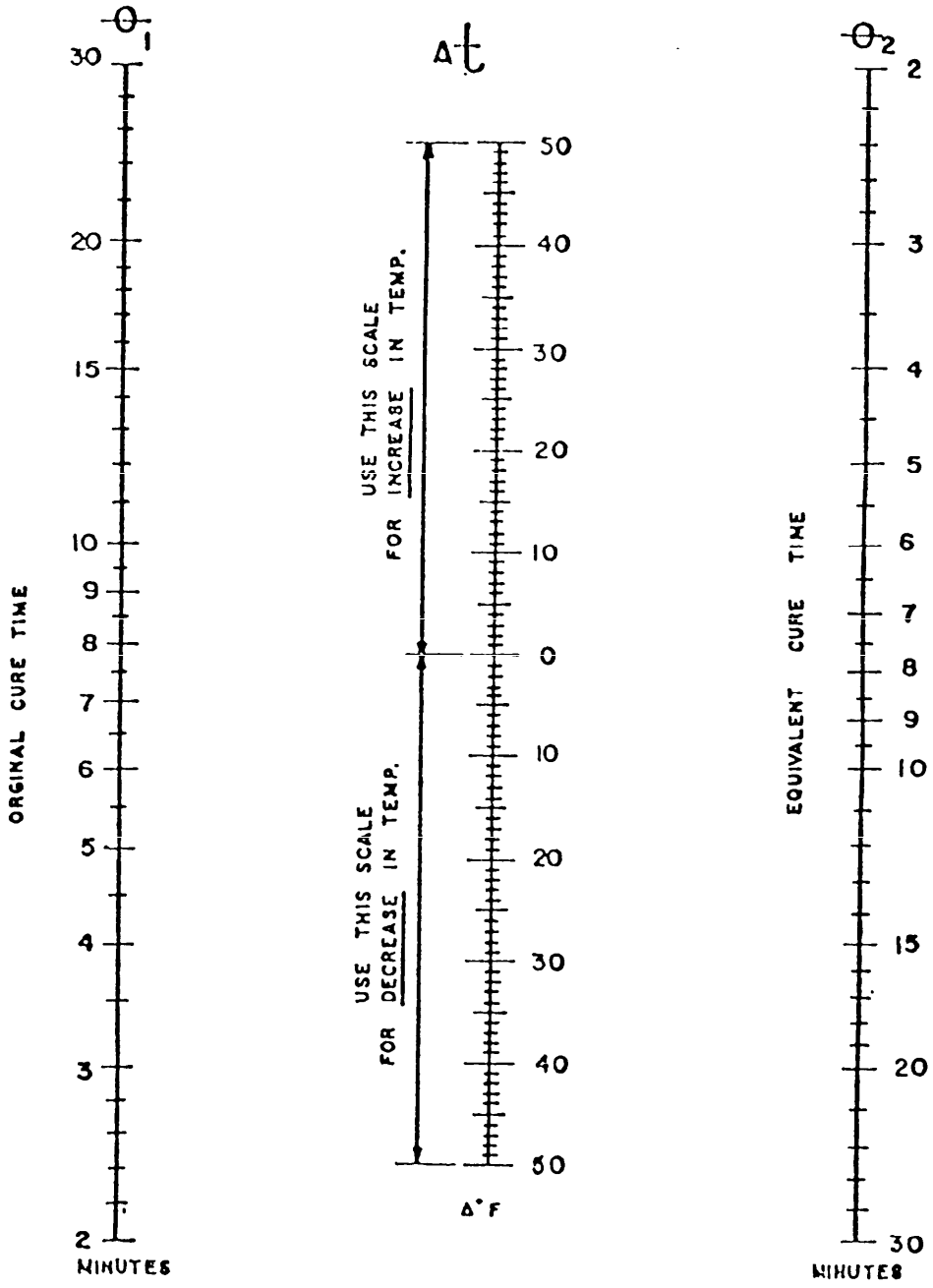


Figure 1.

USING A STRAIGHT EDGE, CONNECT THE ORIGINAL CURE TIME ON θ_1 SCALE WITH THE CHANGE OF CURE TEMPERATURE ON Δt SCALE. EXTEND THE LINE UNTIL IT INTERSECTS θ_2 SCALE AT THE EQUIVALENT CURE TIME.

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6.8 Interplant shipments. Packaging for interplant shipment is for supplies and materials that do not enter the military supply system. Typical interplant shipments are shipments from a vendor to a subcontractor or a prime contractor or from a subcontractor to a prime contractor, or from a vendor or contractor to a military arsenal or plant.

Custodians:

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Preparing activity:

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-F-51286B(EA)	FACEBLANKS, CHEMICAL-BIOLOGICAL MASK, C17R1

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation.

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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